DYSANOS

Generative Dynamic Smooth Arbitrage-free Non-parametric Option Surfaces

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Abstract

This article presents with DYSANOS the first generative market model for smooth SANOS option surfaces for all strikes and expiries which are free of static arbitrage. Our model is designed to generate entire paths of daily spot and option prices for years in the future.

We present a robust and useful if somewhat simplistic baseline in the form of an AR(1) model. We discuss model setup, data pipeline, and training and investigate market reconstruction, stability, and tail behavior. We illustrate model performance on 891 Option Metrics IvyDB S&P Index surfaces from 2022 to 2025, drawn from an underlying archive beginning in 2020. With explicit 1D–4D expiry nodes, the two-η affine-200 model places 80.9% of 3.57 million quotes inside bid–ask; replacing the earlier expiry-specific smoothing multipliers by one front and one term coordinate changes this rate by less than one percentage point. In a 158-date 2025 diagnostic for one-day ATM options, historical-scenario spot hedges reduce normalized root-mean-square P&L by 45–48%; mean–variance, entropy OCE, and power-exponential OCE hedges give similar results. A 159-date open-interest proxy gives a similar conclusion at portfolio scale, reducing RMS P&L from 28.0 to about 14.4 basis points of gross open interest. Finally, a no-replay scalar log-variance diffusion recovers much more of the observed tail mass and tail persistence than Gaussian innovations, although it still underproduces two- and three-week extremes. Block-randomization tests of decoded 2025 surface paths do not reject Gaussian, heavy-tail, diffusion, or historical-replay candidates at the 5% level.

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1 Introduction

The implied volatility surface of an option market changes dynamically over time, giving rise to Vega risk when risk managing option portfolios. Modelling the dynamics of a full continuous surface of options in a way that is simultaneously *statistically realistic* and *statically arbitrage-free* at every simulated date has remained an open problem since the foundational work of [CFD02] on dynamics of the implied volatility surface.

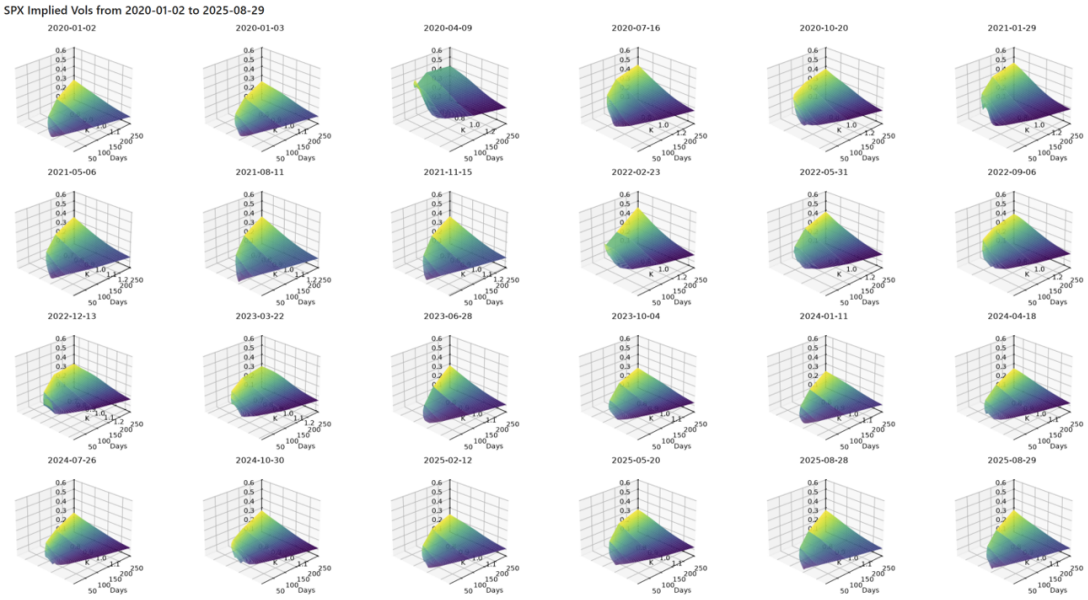


Figure 1: Historical implied volatilities for mid-prices of S&P 500 Index options from 2020-01-02 to 2025-08-29 using Option Metrics Ivy DB data via WRDS `https://wrds-www.wharton.upenn.edu/` (the latest data available at the time of writing).

The most advanced approach until recently were the simulators [WBWB19, WWP+21, CRW23] for realistic dynamics of a *discrete floating grid* of options which are strictly arbitrage-free at each simulation date. These simulators are using, in essence, the discrete local volatility representation pioneered in [BR15] to ensure absence of static arbitrage in every simulated option surface.

However, the restriction of the above models to a discrete set of floating option strikes and time-to-expiries means that the simulators do not have access to the daily returns $dC_t(\tau, k)$ of an actual tradable instruments with cash strikes k and calendar expiries τ (because the contract terms change from one simulated day to another). That means

- Hedging instruments chosen at simulation time determine the hedging instruments available to any hedging algorithm.
- For hedging algorithms such as Deep Hedging [BGTW19] for which we solve optimal trading along full paths, the calculation of the return of a hedging instrument requires the simulation of the spot until the expiry of the farthest option. This limits the expiries of tradable instruments made available during simulation.
- The approach is unsuitable for "daily" hedging algorithms such as Statistical Hedging [Bue17] (a generalization of local regression hedging) or Deep Bellman Hedging [BPW22] (the "actor critic" version of Deep Hedging) which we wish to execute on a simulated path to hedge exotics to expiry – in both cases the P&L of option returns is central to the approach; see also the lecture notes [Bue26].

Of course, since the discrete option prices are free of arbitrage, one option would be to interpolate option prices linearly, which means all options are free of static arbitrage. However, as shown in [Bue06] and illustrated in [BHK+26] linear interpolation yields the most expensive possible arbitrage-free price for any option off the grid, rendering the approach unsuitable in practice. Being linear also means that the implied probability density is piecewise constant and zero between strikes. In other words the spot price can only only take values at the model strikes for each expiry to avoid dynamic arbitrage. This is the reason for the very restrictive and unnatural dynamics in [SW08].

This article presents a model framework for

- **(i) realistic generative dynamics** of
- **(ii) smooth option surfaces for all strikes and expiries** which are
- **(iii) free of static arbitrage**.

This article is a refinement of material presented at The XIII Bachelier World Congress in Bologna.[1]

Historic Context and Literature Review

The classical approach and its limitations. Quantitative finance has traditionally approached this problem by specifying analytically tractable risk-neutral dynamics for the underlying asset — Heston, rough volatility, SABR — and then *calibrating* model parameters to fit each day's option surface independently. In this paradigm, realistic surface dynamics are an afterthought: calibration is repeated from scratch on each date, and inter-day consistency is achieved, if at all, via a local-volatility overlay such as the seminal [GHL12] that lacks interpretable dynamics. This approach is both impractical for large-scale simulation and structurally inconsistent with the goal of generating realistic implied volatility or option price paths.

Why simulating directly in implied-volatility space fails. A natural data-driven alternative is to treat the implied volatility surface itself as the simulation object, much as implied in [CFD02]. The fundamental difficulty is that there is no known parametrization of the implied volatility surface which are free of static arbitrage. If a simulated surface admits a butterfly spread or calendar spread arbitrage, then a learning agent can exploit an artificial profit which is unrelated to the problem for which the generated market was intended. In [CV23], the authors proposed to reduce the probability weights of paths where static arbitage in a simplified form exists; however they do not show that the resulting market is free of arbitrage.

In addition, there are number of recent works such as [NJZB23] which use machine learning methods to learn dynamics of implied volatilities who attempt to impose no-arbitrage via soft constraints; none of them strictly excludes arbitrage. Here, we aim higher and present a framework which does not require minutiae daily guesses on penalty weights and other soft parameters to try avoid static arbitrage along simulated paths of years of implied volatilities.

Also, most published works simplify the problem by only considering artificial expiries. SANOS was shown in [BHK+26] to be able to fit entire option surfaces from one day to expiry out to several years. Our model here is a generative version thereof.

Relation to Dupire's local volatility. An intuitive alternative is to use Dupire's local volatility as state space instead *(not as dynamical model)*. Given a non-negative local volatility function $\sigma_t(T,K)$ for $(T,K) \in \mathbb{R}^2_{\geq 0}$ defined on some day t we can define a smooth, strictly arbitrage-free option surface $C_t(T,K) := C^{\mathrm{LV}}(\sigma_t;T,K)$ by solving the Dupire forward-PDE $\partial_T C^{\mathrm{LV}}(\sigma_t;T,K) = \frac{1}{2}K^2\sigma_t^2(T,K)\,\partial^2_{KK}C^{\mathrm{LV}}(\sigma_t;T,K)$. That means any statistical generative model of Dupire's local volatility gives rise to a dynamic model of option surfaces which are by construction free of static arbitrage.

However, in order to price any simulated option off a simulated local volatility surface we would need to solve a forward-PDE on a suitably fine grid which is slow, particularly when used within batch-based gradient descent training.

The idea to address this conundrum is to use a grid of options and define a discrete version of local volatility on that grid, which is what we called *discrete local volatility (DLV)* in [BR15]. However, its native option prices for strikes not on its base grid are the result of linear interpolation

[1] https://quantitative-research.de/dl/DYSANOS.pdf

and therefore, as mentioned above, "too expensive". We therefore use our smooth SANOS extension [BHK+26].

We also note that in order to find a Dupire local volatility surface for given market data in the first place, we need to fit the market with a smooth arbitrage-free option surface parametrization. Of course, that is what SANOS [BHK+26] was made for but once you have SANOS we might as well use it as the ground truth (since the respective Dupire local volatility will just be another representation of the same surface).

Key contributions

SANOS Decoder:

We show how to implement and learn a coordinate transformation of our SANOS [BHK+26] model such that the state parameters h_t live in $\mathbb{R}^{200}$. A depth-zero affine decoder maps this state into all native DLV, forward-volatility, and forward smoothing coordinates. The call prices remain smooth and statically arbitrage-free.

This "ML-SANOS" is a natural decoder of a driving state process into an option surface of the form

$$h_t \longmapsto C(h_t; T, K) \ .$$

We show how ML-SANOS can be trained directly to suitably pre-processed option market data, yielding a trained h_t for each day. The pre-processing step means that DYSANOS, via SANOS, is fitted to real market bid/ask prices.

Generative Dynamics

We fit a stable 200-dimensional weekly AR(1) directly to the learned surface state. Its transition is parameterized as the matrix exponential of a dissipative generator, and its full innovation covariance is learned by a nonsingular Cholesky factor. The resulting stationary law permits exact Gaussian transitions at daily, weekly, or longer horizons; as non-Gaussian alternatives, a no-replay log-variance diffusion is fitted and historical one-week innovations can be resampled. Although the fitted state has 200 coordinates, decoded implied-volatility variation is strongly concentrated: the first four decoded-IV principal components explain 95.63% for the weekly model and 94.50% for the daily+weekly model.

We do not claim that this approach provides the most realistic dynamics; rather, we see this implementation as baseline to beat with more modern time series models.

Static arbitrage and statistical dynamics

Every decoded SANOS surface is free of static arbitrage by construction. This cross-sectional property does not impose martingale restrictions on fixed-contract returns through simulated time. The present paper studies a historical statistical law and makes no additional dynamic pricing claim.

The completed statistical dynamics and their empirical diagnostics are reported in Section 4.

2 DYSANOS

We consider European call options on a single underlying for historic days $t \in \{t_1, \ldots, t_{n^t}\}$. For notational simplicity we assume forwards, discount factors, and default intensities are zero; the extension to non-zero rates and dividends is standard. We comment in remark 4.1 on how to convert market data into driftless data.

We will use boldface for market observables.

We use $\boldsymbol{S}_t$ to denote the (positive) stock price at time t which we assume trades without cost (and therefore has no bid/ask spread). We assume we observe $L_t \in \mathbb{N}$ European call option prices with time-to-expiries $T_t^\ell \geq 0$ and relative strikes $K_t^\ell \geq 0$ for $\ell = 1, \ldots, L_t$ in the market. Their payoff is therefore

$$\left(\frac{\boldsymbol{S}_{t+T_t^\ell}}{\boldsymbol{S}_t} - K_t^\ell \right)^+ .$$

We denote by $\boldsymbol{B}_t^\ell$ and $\boldsymbol{A}_t^\ell$ their bid and ask prices; mid-price and half-spread are defined (in vector notation) as $\boldsymbol{C}_t := \frac{1}{2}(\boldsymbol{A}_t + \boldsymbol{B}_t)$ and $\boldsymbol{\gamma}_t := \frac{1}{2}(\boldsymbol{A}_t - \boldsymbol{B}_t)$. We generally assume that $\boldsymbol{B}_t \succ 0$ and $\boldsymbol{\gamma}_t \succ 0$.[2] Contrary to most derivative literature, we do not assume that bid/ask prices are in some way free of arbitrage; instead we will show how to handle arbitrage valuations gracefully.

The market Black–Scholes implied mid-volatilities $\boldsymbol{\sigma}_t^\ell$ are given by the unique relation $\mathrm{BSCall}(\boldsymbol{S}_t, K_t^\ell, \boldsymbol{\sigma}_t^\ell \sqrt{T_t^\ell}) = \boldsymbol{C}_t^\ell$ in terms of the Black & Scholes call price formula

$$\mathrm{BSCall}(S, K, W) := S\mathcal{N}(d_+) - K\mathcal{N}(d_-) \quad \text{for} \quad d_\pm := \frac{\log S/K}{W} - \frac{1}{2}W^2 \,. \tag{1}$$

where $\mathcal{N}$ denotes the normal distribution function.

2.1 Arbitrage-Free Call Price Functions

A call price function is simply a function $C : \mathbb{R}^2_{\geq} \longrightarrow \mathbb{R}_{\geq 0}$ which computes call prices $C(T, K)$ for time-to-expiries $T \geq 0$ and relative strikes $K \geq 0$ (i.e. the call prices are normalized to a spot price of 1). The objective of this section and the next is to generate such functions C_t for each simulated date t such that each C_t is "free of static arbitrage" in the following sense:

Definition 2.1. *A call price surface $C : \mathbb{R}^2_{\geq 0} \to \mathbb{R}_{\geq 0}$ is* statically arbitrage-free *or* free of static arbitrage *if there is no trading strategy which has non-negative returns with a non-zero probability of positive returns.*

Theorem 2.2. *A call price surface C is free of static arbitrage if and only if it does not admit negative butterfly or calendar spreads and satisfies the boundary conditions $C(T, 0) = 1$, $C(0, K) = (1 - K)^+$, $\lim_{K \uparrow \infty} C(T, K) = 0$ and $\partial_K C(T, 0) \in [-1, 0)$.*

The stricter condition $\partial_K C(T, 0) = -1$ applies if the stock price cannot become zero.

This has been proven in [BR15]. Here we give a practical guide on how to construct *actionable* arbitrage strategies if any of the above are violated. The simplicity of each of them illustrates the risk of admitting arbitrage in generative environments which are used to train agents to hedge: if such opportunities exist, an optimal agent will try learning them.

Remark 2.3 (Actionable arbitrage if the conditions of theorem 2.2 are not met)**.** *All of the below are meant as practical heuristics.*

- ***Butterfly arbitrage:*** *this means that the price of a positive butterfly payoff is negative. The arbitrage is to get paid to enter into the butterfly which can only ever generate a positive return.*
- ***Calendar arbitrage:*** *if for the same strike K the call price at T_2 is below the call price at T_1, then:*

  1. *Sell the call at T_1 and buy the call at T_2. This provides a positive cash flow $\pi_0 > 0$.*

[2] We define $x \succ y \Leftrightarrow \forall i : x^i > y^i$.

2. *In T_1 if the call expiries out of the money, then we have made money: both π_0 and a potential positive payoff at T_2 in our favour.*

   If the call expires in the money, then we deliver one share and receive K. We are now short that share. In T_2, if the call expiries in the money, then we receive one share to cover our short in return for K. Our total cashflow is π_0. If the call expiries out of the money, then $S_T < K$. We can therefore buy back the share and cover our short for less than K. Our profits are $\pi_0 + (K - S_T)^+ > 0$.

- *If $C(0,K) \neq (1-K)^+$ we have an immediate pathological arbitrage.*
- *If $C(T,0) > 1$ then sell the call and cover with one share; if $C(T,0) < 1$ then buy the call and enter into a short equity position.*
- *$\partial_K C(T,0) \geq -1$: assume on the contrary there is a $K \downarrow 0$ such that $(1-C(T,K))/(-K) < -1$ i.e. $C(T,K) = 1-K-\epsilon$ for some $\epsilon > 0$. Buy the call and short sell the equity, resulting a cash position of $\pi_0 = -K + \epsilon$; the payoff in T is $(S_T - K)^+$ which allows us to cover our position and cash with a total reward of $\epsilon > 0$.*
- *$\partial_K C(T,0) < 0$: assume on the contrary there is a $K \downarrow 0$ such that $(1-C(T,K))/(-K) \geq 0$. We show existence of arbitrage in the case of equality i.e. existence of some $K > 0$ such that $C(T,K) = 1$. In that case we sell the call and buy the equity for 1. If the call expires in the money, we deliver the share and receive $K > 0$. If the call expiries out of the money, then sell the share for $S_T \geq 0$.*
- *$\lim_{K\uparrow\infty} C(T,K) = 0$ which means for all $c > 0$ there exists some K such that $C(T,K') < c$ for all $K' \geq K$ (because of convexity). This is the only arbitrage condition which is technical because a violation means, essentially, that the equity can grow without bounds which violates integrability. Assume therefore on the contrary that there is some $c > 0$ such that $C(T,K) \geq c$ for all K. Notice that, formally,*

$$S_T = -\int_0^\infty \frac{(S_T - K + dK)^+ - (S_T - K)^+}{dK} dK = S_T + \lim_{K\uparrow\infty} (S_T - K)^+ .$$

Therefore sell (a discretization of) the strip

$$\int_0^\infty \frac{C(T,K) - C(T,K+dK)}{dK} dK = 1 + e$$

which delivers S_T, and buy the equity. This will generate profits of $e > 0$.

2.2 SANOS - Smooth Arbitrage-free Non-parametric Option Surfaces

In [BHK+26] we introduced a non-parametric family of call price functions which are all statically arbitrage-free. For clarity, let us briefly fix t and drop it from the notation of market data and model parameters.

Each call price function is given in terms of model expiries $0 = \hat{T}^0 < \hat{T}^1 < \cdots < \hat{T}^M$ and, for each model expiry, $N^j > 0$ inner strikes. Including the two boundary strikes, write $\hat{K}^j = (\hat{K}^{j,i})_{i\in\mathcal{I}^j}$ with $\mathcal{I}^j := \{0,\dots,N^j+1\}$ and $0 < \hat{K}^{j,0} < \cdots < 1 < \cdots < \hat{K}^{j,N^j+1}$. It is recommended that the two outer strikes are far away from the inner strikes.[3] Model strikes and expiries do not need to match market strikes or expiries. We also set $\mathcal{I}^0 = \{0\}$, $\hat{K}^{0,0} = 1$, and $q^{0,0} = 1$; see [BHK+26] for general comments on grid construction.

[3] In our experiments, the inner strikes $0 < \hat{K}^{j,1} < \cdots < 1 < \cdots < \hat{K}^{j,N^j}$ were chosen first and then extended by $\hat{K}^{j,0} := \hat{K}^{j,1}/2$ and $\hat{K}^{j,N^j+1} = 1.5\,\hat{K}^{j,N^j}$.

We say that $q = (q^1, \dots, q^M)$ with $q^j \in \mathbb{R}^{N^j+2}$ is a *martingale density* if it satisfies the linear constraints

$$\begin{cases} \text{Each } q^j \text{ is a density:} & q^j \geq 0,\ (q^j)'1 = 1 \\ \text{Each } q^j \text{ has unit expectation:} & (q^j)'\hat{K}^j = 1 \\ \text{The } q \text{ are in convex order:} & c^j \geq c^{j|j-1} \end{cases} \tag{2}$$

where $c^{j,i} = q^{j\prime}(\hat{K}^j - \hat{K}^{j,i})^+$ is the j-call price for $\hat{K}^{j,i}$, and $c^{j|j-1,i} := q^{j-1\prime}(\hat{K}^{j-1} - \hat{K}^{j,i})^+$ is the $(j-1)$-call price for $\hat{K}^{j,i}$.

Definition 2.4 (SANOS call price)**.** *Let q be a martingale density and $0 < W^1 < \cdots < W^M$ be increasing smoothing total volatilities.*[4] *The SANOS call price for arbitrary $T \geq 0, K \geq 0$ is given by*

$$\begin{aligned} C(T,K) := \alpha^j(T) \sum_{i \in \mathcal{I}^j} \mathrm{BSCall}(\hat{K}^{j,i}, K, W(T)) q^{j,i} \\ + (1 - \alpha^j(T)) \sum_{i \in \mathcal{I}^{j-1}} \mathrm{BSCall}(\hat{K}^{j-1,i}, K, W(T)) q^{j-1,i}, \qquad T \in (\hat{T}^{j-1}, \hat{T}^j]. \end{aligned} \tag{3}$$

with $\alpha^j(T) = (T - \hat{T}^{j-1})/(\hat{T}^j - \hat{T}^{j-1})$ and total volatility $W(T) := \alpha^j(T)W^j + (1 - \alpha^j(T))W^{j-1}$. For $T > \hat{T}^M$ we set $\alpha^{M+1}(T) = 0$ and $W(T) := W^M\sqrt{T/\hat{T}^M}$; however extrapolation beyond the last model expiry is generally not recommended.

Theorem 2.5. *The call price function C is statically arbitrage-free. For $W(T) > 0$ it is C^∞ in K. The function is as smooth as α in T.*

Remark 2.6. *Contrary to [BHK+26] we interpolated the Black & Scholes backbone in total volatility linearly, not variance. This is a mechanical choice to avoid taking square roots of variables during training which can lead to exploding gradients. If forward volatilities are bounded well away from zero, linear interpolation in variance is suitable, too.*

Remark 2.7. *The Black & Scholes formula in (3) can be replaced by the call price function for any martingale in T, for example the FFT Heston price formula [CM99]. For practical implementations, the function should be fast on a GPU.*

Calibration as a linear program: We assume we obtain $W^1 < \cdots < W^M$ from the market e.g. as square roots of ATM variance per expiry. Given W the model call price $C^\ell := C(T^\ell, K^\ell)$ for a market expiry T^ℓ and a market strike K^ℓ can be seen in (3) to be linear in q. The market-fitting problem is therefore the linear programme

$$\min_q \sum_\ell \frac{1}{\boldsymbol{\gamma}^\ell} [(C^\ell - \boldsymbol{A}^\ell)^+ + (\boldsymbol{B}^\ell - C^\ell)^+ + \epsilon |C^\ell - \boldsymbol{M}^\ell|] \tag{4}$$

subject to the linear constraints (2) encoding non-negativity, normalisation, integration to 1, and the inter-expiry convex-order constraints that enforce absence of calendar-spread arbitrage. We note that the model strikes and time-to-expires do not need to correspond to market strikes or expiries.

On S&P 500 Index data with 1000 options across 48 expiries (2-day to 657 business-day TTE), the LP solves in sub-seconds on a desktop PC, achieving 91.4% of options within bid–ask on a representative date (2025-05-06), with a median error of 21% of the half-spread on the remaining options. Figure 2 illustrates the quality of the fit. The example is from [BHK+26]; refer to that work for further details.

[4] The linear-programming preprocessing uses 50% of ATM total volatility. The trained ML-SANOS decoder instead learns the positive forward smoothing multipliers η^j.

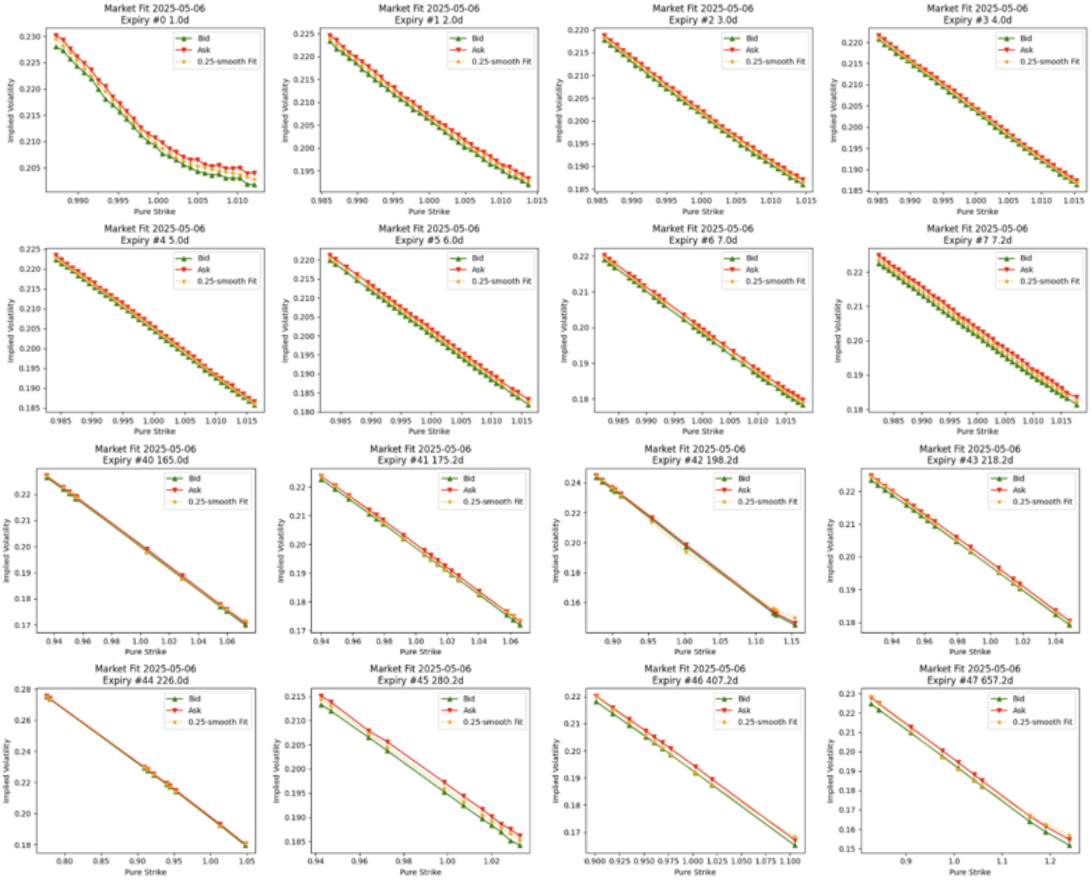


Figure 2: Example fit of SANOS.

3 ML-SANOS

We may view our call price operator as a map from the model parameters to a function space

$$(q, W;\, \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C \tag{5}$$

for which we naturally want to learn q, W and, as we will see, $\hat{K}$ (as the right strike grid depends on the implied variance of the market). However, the density weights q^j in particular are subject to the linear constraints (2) for which no exact methods exist to integrate them into standard batch gradient descent methods. In this section we therefore reparameterize the map (5) in terms of learnable parameters.

To this end, recall the previously defined linear call prices $c^{j,i} := (q^j)'(\hat{K}^j - \hat{K}^{j,i})^+$ and $c^{j|j-1,i} := (q^{j-1})'(\hat{K}^{j-1} - \hat{K}^{j,i})^+$. Given a martingale density q the squared discrete local volatilities (DLVs) $\Sigma^{j,i} \geq 0$ were then defined for the inner indices $i = 1, \dots, N^j$ in [BR15] via the discrete "backward" Dupire formula[5]

$$\frac{c^{j,i} - c^{j|j-1,i}}{\hat{T}^j - \hat{T}^{j-1}} \;\equiv\; \Sigma^{j,i} \frac{(\hat{K}^{j,i})^2}{(\hat{K}^{j,i+1} - \hat{K}^{j,i})(\hat{K}^{j,i} - \hat{K}^{j,i-1})}\, q^{j,i} \tag{6}$$

There is no DLV coordinate at the two boundary indices $i = 0$ and $i = N^j + 1$; we set $\Sigma^{j,i} = 0$ whenever $q^{j,i} = 0$.

The key is that we can reverse this representation. We briefly recall the construction from [BHK+26]:

[5]This actually *is* the Dupire formula applied to the implicit forward PDE of the density of a local volatility process, correctly computed on an irregular strike grid; c.f. [BR15].

1. Start in $q^0 := (1)$ for $\hat{K}^0 := (1)$.

2. For $j > 0$ if $\hat{K}^j = \hat{K}^{j-1}$; then $q^{j|j-1} = q^{j-1}$.

   If this is not the case we construct $q^{j|j-1}$ as follows: let $c^{j|j-1,i} := \sum_{\ell \in \mathcal{I}^{j-1}} q^{j-1,\ell}(\hat{K}^{j-1,\ell} - \hat{K}^{j,i})^+$ for $i \in \mathcal{I}^j$. Augment these call values by $\hat{K}^{j,-1} = 0$ and $c^{j|j-1,-1} = 1$, and define

   $$dC^{j|j-1,i} := \frac{c^{j|j-1,i} - c^{j|j-1,i-1}}{\hat{K}^{j,i} - \hat{K}^{j,i-1}} \qquad (i \in \mathcal{I}^j).$$

   With $dC^{j|j-1,N^j+2} = 0$, define for $i \in \mathcal{I}^j$

   $$q^{j|j-1,i} := dC^{j|j-1,i+1} - dC^{j|j-1,i} \ .$$

   Then $q^{j|j-1}$ is a density over $\hat{K}^j$, given by linear interpolation of call prices at the previous expiry. In [BHK+26] this operation was written as the linear expression

   $$q^{j|j-1} = L^j q^{j-1}$$

   in terms of the respective matrix $L^j \in \mathbb{R}^{|\mathcal{I}^j| \times |\mathcal{I}^{j-1}|}$.

3. Let now

   $$\xi^{j,i\pm} := \pm \frac{(\hat{K}^{j,i})^2}{(\hat{K}^{j,i+1} - \hat{K}^{j,i-1})\,(\hat{K}^{j,i\pm1} - \hat{K}^{j,i})} \Sigma^{j,i}(\hat{T}^j - \hat{T}^{j-1}) \geq 0$$

   for $i = 1, \ldots, N^j$ which we note is linear in $\Sigma^{j,i}$. The discrete local volatility transition operator is defined by the (inverse of) the tri-band matrix $\Xi^j \in \mathbb{R}^{(N^j+2)\times(N^j+2)}$ with

   - upper diagonal $(-\xi^{j,1-}, \ldots, -\xi^{j,N^j-}, 0)$,
   - lower diagonal $(0, -\xi^{j,1+}, \ldots, -\xi^{j,N^j+})$, and
   - main diagonal $(1, 1 + \xi^{j,1+} + \xi^{j,1-}, \ldots, 1 + \xi^{j,N^j+} + \xi^{j,N^j-}, 1)$.

   (We note that Ξ^j is linear in the discrete local volatilities $\Sigma^{j,i}$.)

   As shown in [BR15] the inverse $(\Xi^j)^{-1}$ exists, and can be efficiently computed on GPU using a compiled Thomas' algorithm. It is also a martingale transition operator in the sense of [BR15], and is used to define the linear operation

   $$q^j := (\Xi^j)^{-1} q^{j|j-1} \ .$$

   The operator Ξ^j is the in-homogeneous strike implicit transition operator of the PDE implied by (6), and is based on ideas of [AH11].

With the above we have established a map from much simpler non-negative discrete local vols squared to densities, and then to SANOS call prices:

$$(\Sigma, W;\, \hat{K}, \hat{T}) \overset{\text{DLV}}{\longmapsto} (q, W;\, \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C \ . \tag{7}$$

That allows us to pass to a practically unconstrained parameter set, which we call the ML-SANOS parametrization. Recall function $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$.

Definition 3.1 (ML-SANOS)**.** *Fix the normalized grid geometry for ML-SANOS with:*

- *Time-to-expiries (TTEs)* $0 = \hat{T}^0 < \cdots < \hat{T}^M$ *with* $\Delta\hat{T}^j = \hat{T}^j - \hat{T}^{j-1}$*;*
- *Normalized strikes* $z^1 < \cdots < 0 < \cdots < z^N$ *which we assume contain zero;*
- *Minimum and maximum forward volatilities* $\sigma_{\max} > \sigma_{\min} > 0$*;*

- *Minimum and maximum squared discrete local volatilities* $\Sigma_{\max} > \Sigma_{\min} > 0$*; and*
- *Minimum and maximum smoothing factors* $1 > \eta_{\max} > \eta_{\min} \geq 0$*.*

We translate the new real-valued ML-SANOS parameters $u = (u^{\Sigma}, u^{\sigma}, u^{\eta,\mathrm{f}}, u^{\eta,\mathrm{term}}) \in \mathbb{R}^{M\times N} \times \mathbb{R}^M \times \mathbb{R}^2$ *into SANOS parameters as follows:*

- *Total volatilities: define the* ***forward volatilities***

$$\sigma^j := \sigma_{\min} + \mathrm{sigmoid}(u^{\sigma,j})(\sigma_{\max} - \sigma_{\min}) \tag{8}$$

and two smoothing factors

$$\begin{aligned} \eta^{\mathrm{f}} &:= \eta_{\min} + (\eta_{\max} - \eta_{\min})\mathrm{sigmoid}(u^{\eta,\mathrm{f}}), \\ \eta^{\mathrm{term}} &:= \eta_{\min} + (\eta_{\max} - \eta_{\min})\mathrm{sigmoid}(u^{\eta,\mathrm{term}}), \end{aligned}$$

where the interval multiplier is $\eta^1 := \eta^{\mathrm{f}}$ *and* $\eta^j := \eta^{\mathrm{term}}$ *for* $j = 2, \ldots, M$*. Thus the front interval can set the short-end smoothing scale, while a single term parameter controls every subsequent increment. With these quantities define the* ***total volatilities***

$$\widetilde{W}^j := \sum_{\ell=1}^{j} \sigma^\ell \sqrt{\Delta\hat{T}^\ell} \qquad \textit{and} \qquad W^j := \sum_{\ell=1}^{j} \eta^\ell \sigma^\ell \sqrt{\Delta\hat{T}^\ell}\ . \tag{9}$$

- *We use* $\widetilde{W}^j$ *to define the* ***relative strikes***

$$\hat{K}^{j,i} := \exp\left(z^i \widetilde{W}^j\right)\ . \tag{10}$$

with boundaries $\hat{K}^{j,0} := \frac{1}{2}\hat{K}^{j,1}$ *and* $\hat{K}^{j,N+1} := \frac{3}{2}\hat{K}^{j,N}$*.*

- *Finally, we use the new real-valued ML-SANOS parameters to define the* ***squared discrete local volatilities****:*

$$\Sigma^{j,i} := \min\left\{\Sigma_{\max}, \Sigma_{\min} + \exp(u^{\Sigma,j,i})\right\}\ . \tag{11}$$

The raw log-DLV coordinate therefore has no saturating upper sigmoid. During fitting, a smooth one-sided barrier supplies a restoring gradient as it approaches the pricing cap $\Sigma_{\max}$*.*

Put together, this process describes a map

$$(\,u; z, \hat{T}\,) \overset{\text{ML SANOS}}{\longmapsto} (\,\Sigma, W; \hat{K}, \hat{T}) \overset{\text{DLV}}{\longmapsto} (\,q, W; \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C\ . \tag{12}$$

In our new parametrization u *lives in an unrestricted space and is therefore particularly suited for ML/AI based learning. The expiry locations remain fixed.*

3.1 Learning ML-SANOS from real data

The completed experiments use the following transparent affine specialization of the unrestricted ML-SANOS coordinates introduced above.

3.1.1 Affine-200 specialization

The reported models retain the unrestricted-coordinate idea above but make the compression map deliberately transparent. Let $N = 200$ be the number of inner support strikes and M the number of model expiries. For each date the uncompressed state is stored as an expiry-major DLV/forward-volatility block followed by the two smoothing coordinates:

$$u_t = \Big((u_t^{\Sigma,j,1}, \dots, u_t^{\Sigma,j,N}, u_t^{\sigma,j})_{j=1}^{M}, u_t^{\eta,\mathrm{f}}, u_t^{\eta,\mathrm{term}} \Big) \in \mathbb{R}^{M(N+1)+2}. \tag{13}$$

The completed decoder has no hidden layers:

$$u_t = b + Bh_t, \qquad h_t \in \mathbb{R}^{200}, \tag{14}$$

where $b \in \mathbb{R}^{M(N+1)+2}$ and $B \in \mathbb{R}^{(M(N+1)+2)\times 200}$. The weekly model has $M = 14$ and therefore 2,816 native coordinates; the daily+weekly model has $M = 18$ and 3,620 native coordinates. The 200 coordinates of h_t are learned latent coordinates, not principal components of implied volatility.

The choice of 200 is an empirical capacity compromise rather than a structural claim about the intrinsic dimension of option surfaces. It is large enough to preserve almost all of the direct native quote fit while remaining below the number of historical dates, so a full innovation covariance and stable joint state dynamics can still be estimated. We report the native-versus-200 fit loss explicitly below rather than treating this dimension as exact. The dimension was fixed before, and was not selected using, the 2025 tail, path, or hedging diagnostics.

For initialization from an expiry-specific smoothing curve, the front coordinate is copied directly. The shared term coordinate is the forward-total-volatility-weighted mean of the later multipliers. This choice preserves the final-expiry value of W_t^M exactly before the two-coordinate model is re-optimized; it is an initialization identity, not a restriction on the subsequent fit.

The native coordinates are converted into the positive SANOS parameters by

- $\Sigma_t^{j,i} = \min\Big\{\Sigma_{\max}, \Sigma_{\min} + e^{u_t^{\Sigma,j,i}}\Big\}$;
- $\sigma_t^j = \sigma_{\min} + (\sigma_{\max} - \sigma_{\min})\mathrm{sigmoid}(u_t^{\sigma,j})$; and
- η_t^{f} and η_t^{term} are the bounded sigmoid transforms above, with $\eta_t^1 = \eta_t^{\mathrm{f}}$ and $\eta_t^j = \eta_t^{\mathrm{term}}$ for $j \geq 2$.

Here $\Sigma_t^{j,i}$ is the squared DLV, σ_t^j is the forward volatility, and η_t^j is its smoothing-backbone multiplier. The experiment uses $(\Sigma_{\min}, \Sigma_{\max}) = (0.01^2, 3^2)$, $(\sigma_{\min}, \sigma_{\max}) = (0.01, 1)$, and $(\eta_{\min}, \eta_{\max}) = (0.01, 0.50)$. A smooth cap penalty acts on the raw log-DLV coordinate without imposing a saturating upper sigmoid. The two cumulative total-volatility curves are exactly

$$\widetilde{W}_t^j = \sum_{\ell=1}^{j} \sigma_t^\ell \sqrt{\Delta \hat{T}^\ell}, \qquad W_t^j = \sum_{\ell=1}^{j} \eta_t^\ell \sigma_t^\ell \sqrt{\Delta \hat{T}^\ell}. \tag{15}$$

Since both σ_t^j and η_t^j are positive, W_t^j is increasing in expiry.

Each date uses one floating relative-strike support shared across all model expiries. Half of the 200 normalized-moneyness candidates cover the front expiry and half the final expiry over $(-3, 2)$; after conversion with the date's total-volatility curve they are merged, sorted, and separated by a minimum log-strike gap. The outer support nodes are one half of the first inner strike and one and a half times the final inner strike. The shared support makes the DLV recursion a sequence of differentiable tridiagonal solves without an expiry-to-expiry regridding operator.

For quote ℓ on date t, let $\widehat{C}_t^\ell$ be the decoded option price, $M_t^\ell = (\boldsymbol{A}_t^\ell + \boldsymbol{B}_t^\ell)/2$, and $\gamma_t^\ell = (\boldsymbol{A}_t^\ell - \boldsymbol{B}_t^\ell)/2$. Decoder fitting minimizes the equal-date-weighted objective

$$\mathcal{L}_t^{\text{quote}} = \frac{1}{L_t}\sum_{\ell=1}^{L_t}\frac{1}{\gamma_t^\ell}\left\{(\widehat{C}_t^\ell - \boldsymbol{A}_t^\ell)^+ + (\boldsymbol{B}_t^\ell - \widehat{C}_t^\ell)^+ + 0.05|\widehat{C}_t^\ell - M_t^\ell|\right\} \tag{16}$$

The native controls first fit an independent u_t on every date. The affine-200 model is initialized from those controls and jointly updates b, B, the historical states h_t, and the stable weekly dynamics in Section 4. A gauge penalty fixes the affine non-identifiability by keeping the fitted states at zero sample mean and unit sample covariance.

3.1.2 Data and model geometries

The empirical population contains 891 SPX dates from 2022-01-03 through 2025-08-29. We exclude the 0.2-business-day AM-settled population and retain 3,566,742 call or put quote observations for the weekly geometry and 3,566,258 for the daily+weekly geometry. Calls and puts at the same expiry and strike count as separate observations.

Both models use 200 inner strikes over normalized moneyness $z \in (-3, 2)$. They differ only in the internal expiry geometry:

$$\begin{aligned} \text{weekly:} &\quad (5, 10, 15, \ldots, 70) \text{ business days,} \\ \text{daily+weekly:} &\quad (1, 2, 3, 4, 5, 10, 15, \ldots, 70) \text{ business days.} \end{aligned}$$

For each geometry we retain a native high-dimensional fit and an affine-200 fit. The native fits isolate the cost of state compression; comparison across geometries isolates the value of explicit short-end expiry nodes.

Inside-spread statistic. For fitted price $\widehat{C}$, bid B, and ask A, define excess in half-spread units by

$$e(\widehat{C}; B, A) := \frac{(\widehat{C} - A)^+ + (B - \widehat{C})^+}{(A - B)/2}. \tag{17}$$

Thus $e = 0$ exactly when the model price lies inside bid–ask. Conditional excess statistics below are computed only over observations with $e > 0$.

3.1.3 Market fit

Table 1 gives the complete quote-population results from the two-η rerun. The corresponding expiry-specific-η presentation results are retained as a like-for-like reproducibility benchmark below.

Model	Quotes	Inside bid–ask	Mean excess	Cond. mean	Cond. median
Weekly native	3,566,742	73.2678%	1.1101	4.1525	1.4261
Weekly affine-200	3,566,742	72.8728%	1.1017	4.0612	1.3925
Daily+weekly native	3,566,258	82.2809%	0.3120	1.7609	0.6920
Daily+weekly affine-200	3,566,258	**80.9211%**	0.3229	1.6924	0.6776

Table 1: Aggregate fit. Excess is defined by (17).

Table 2 compares the rerun directly with the expiry-specific-η artifacts used for the presentation. The same quote population and objective are used in every row. All inside-spread changes are below one percentage point and all changes in mean excess are below 3%, so collapsing the weakly identified multiplier curve to a front and a term coordinate does not explain the model's fit.

Geometry	Mode	Inside: old → two-η	Change	Mean excess: old → two-η
Weekly	Native	73.2908% → 73.2678%	−0.0230 pp	1.1091 → 1.1101
Weekly	Affine-200	72.9723% → 72.8728%	−0.0995 pp	1.0996 → 1.1017
Daily+weekly	Native	81.8429% → 82.2809%	+0.4380 pp	0.3157 → 0.3120
Daily+weekly	Affine-200	80.0138% → 80.9211%	+0.9072 pp	0.3320 → 0.3229

Table 2: Reproducibility of the two-η rerun against the previously reported expiry-specific-η fits.

Affine compression costs only 0.39 percentage points for the weekly geometry and 1.36 percentage points for the daily+weekly geometry. By contrast, adding the four short-end nodes improves affine-200 fit by 8.05 percentage points. Expiry geometry, not latent-state capacity, is therefore the main bottleneck.

The mechanism is visible directly in table 3.

Model	1D	2D	3D	4D	5D
Weekly native	24.07%	32.47%	42.04%	55.66%	45.45%
Weekly affine-200	24.35%	32.65%	42.11%	55.26%	45.26%
Daily+weekly native	94.56%	95.21%	95.57%	95.77%	68.43%
Daily+weekly affine-200	88.59%	91.98%	93.68%	94.13%	67.58%

Table 3: Inside-spread rates at the five shortest nominal expiries.

The weekly model has no independent 1D–4D state directions and must extrapolate from 5D. The daily+weekly model represents front-end dislocations explicitly.

Figure 3 shows that this is a localized geometry effect, not a general loss of fit. The explicit daily nodes lift the 1D–4D inside-spread rate from roughly 24–55% to 89–94% in the affine model. Once the market expiry is comfortably beyond the first weekly node, the two geometries converge and both approach near-complete containment at the longer end. Within either geometry, the native and affine-200 curves remain close; compression is therefore secondary to where the decoder places its expiry nodes.

The same conclusion appears in the magnitude of misses. Conditional on a quote falling outside the spread, Figure 4 measures its distance in half-spread units. A weekly decoder forced to extrapolate before 5D can miss 1D and 2D quotes by more than ten half-spreads on average; explicit daily nodes reduce that short-end error to roughly one to three half-spreads. The gap narrows rapidly after the first week.

Figure 5 gives a cross-sectional view on 2024-06-14. At representative expiries of 10, 23, 43, and 67 business days, the independently fitted market SANOS surface and the completed weekly affine model track one another closely over the central liquid strike range. The longer-expiry panels also expose a material discrepancy at the far-left boundary; the native-versus-affine IV RMSE is 5.30 and 6.87 volatility points there, compared with 0.12 and 0.30 at 10 and 23 days. The example therefore supports the aggregate fit result while making clear that extrapolative wing coordinates remain a limitation.

Interpretation of the 80% number. The 80.92% headline is the affine-compression reconstruction test over the full 891-date population. After joint fitting of the stable state dynamics, the simulation checkpoint places 81.48% of all quotes inside bid–ask; its 2025 slice contains 707,630 observations and has an 80.56% inside-spread rate. The shared decoder and dynamics used all dates, so the latter is a chronological slice rather than a strict temporal holdout. We interpret it as evidence that the fit is not concentrated in the earlier sample, not as an untouched forecasting estimate.

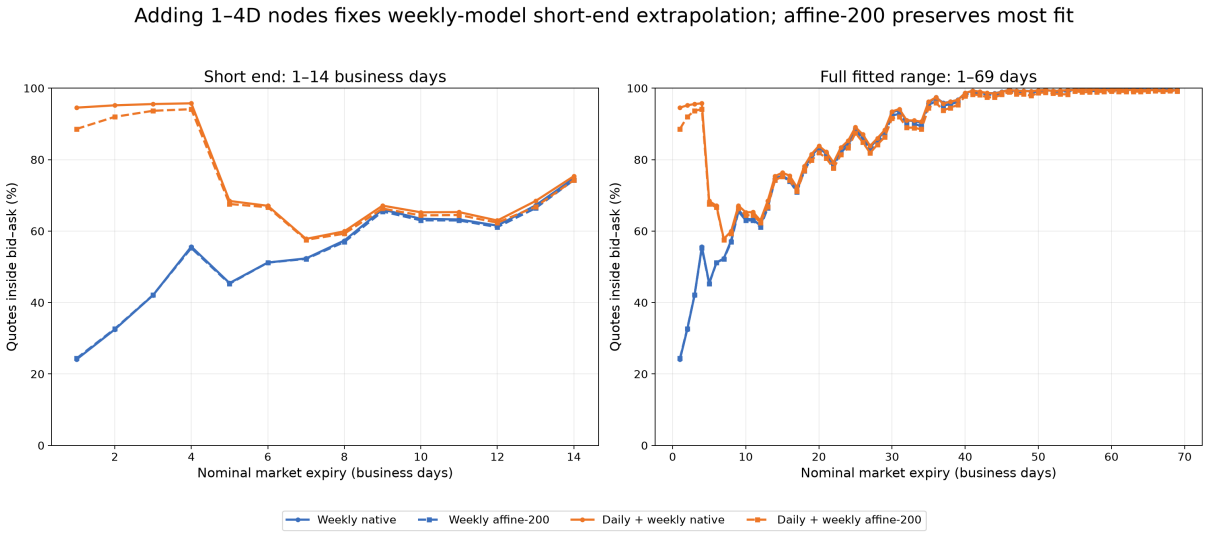


Figure 3: Inside-spread rate by nominal market expiry for the weekly and daily+weekly geometries, before and after affine-200 compression. All quote observations in each expiry bucket are included.

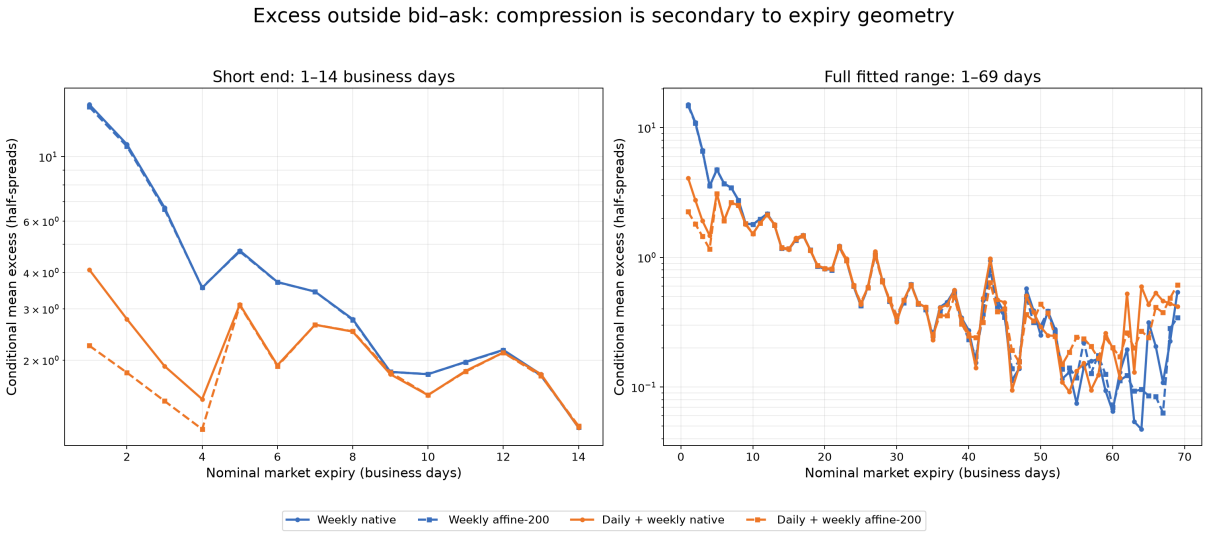


Figure 4: Mean half-spread excess by market expiry, conditional on the model price being outside bid–ask. The logarithmic vertical scale exposes the very large short-end extrapolation errors of the weekly geometry.

4 Statistical option-surface state-space model

4.1 Weekly AR(1)

Let h_n be the affine SANOS state sampled at weekly intervals. The fitted statistical law is

$$h_{n+1} = m + A(h_n - m) + \varepsilon_{n+1}, \qquad \mathbb{E}_{\mathbb{P}}[\varepsilon_{n+1} \mid h_n] = 0, \qquad \mathrm{Cov}(\varepsilon_{n+1}) = Q. \tag{18}$$

Rather than fitting an unrestricted discrete transition and checking it after the fact, the model parameterizes

$$\kappa = LL' + S + \kappa_{\min} I, \qquad S' = -S, \qquad A = e^{-\kappa}, \qquad Q = GG', \tag{19}$$

where L is lower triangular, S is unrestricted skew-symmetric, G is lower triangular with strictly positive diagonal, and $\kappa_{\min} = 10^{-3}$ per week. The symmetric part of κ is strictly positive, so every fitted A is stable throughout optimization. The state therefore has a finite invariant covariance P satisfying

$$P = APA' + Q. \tag{20}$$

In Gaussian mode we obtain exact fractional-week transitions. With $A_\Delta = e^{-\kappa\Delta}$,

$$h_{t+\Delta} = m + A_\Delta(h_t - m) + \varepsilon_\Delta, \qquad \varepsilon_\Delta \sim \mathcal{N}(0, Q_\Delta), \qquad Q_\Delta = P - A_\Delta P A_\Delta'. \tag{21}$$

This avoids Euler instability and permits daily, weekly, or longer Gaussian simulation steps from the same fitted weekly law.

4.2 Innovation laws and tail dynamics

Gaussian innovations are a convenient baseline but miss market tails. As a non-parametric comparator we also simulate at weekly steps by independently resampling centered fitted one-week residuals whose endpoints precede 2025. At two and three weeks, independently drawn residuals are propagated through A and summed; no multi-week block is replayed in this tail experiment.

For an exact-horizon state return r with fitted horizon covariance Q^H, define its normalized squared radius

$$(\delta^H)^2 := \frac{r'(Q^H)^{-1} r}{200}. \tag{22}$$

The Gaussian reference would concentrate this quantity sharply around one in 200 dimensions. The observed radii are both heavier-tailed and serially dependent.

4.2.1 No-replay stochastic-volatility diffusion

We therefore add a scalar latent log-variance diffusion without replaying any historical residual. Keeping the affine conditional mean and loading G from $Q = GG'$, the exact weekly transition is

$$\varepsilon_{n+1} = \sqrt{\rho \exp(g_n - v_g/2)}\, G z_{n+1}, \qquad z_{n+1} \sim \mathcal{N}(0, I), \tag{23}$$

$$g_{n+1} = \phi g_n + \sqrt{v_g(1-\phi^2)}\, \xi_{n+1}, \qquad \xi_{n+1} \sim \mathcal{N}(0, 1), \tag{24}$$

with independent z_{n+1} and ξ_{n+1}. Stationarity gives $g_n \sim \mathcal{N}(0, v_g)$ and hence $\mathbb{E}[\rho \exp(g_n - v_g/2)] = \rho$. Equation (24) is the exact discrete transition of a stationary Ornstein–Uhlenbeck diffusion; it therefore permits fractional-time simulation without Euler discretization.

The parameters (ρ, ϕ, v_g) are fitted on the 703 pre-2025 one-week innovations. Since

$$\log(\delta_n^2) = \log \rho + g_n - v_g/2 + \log(\chi^2_{200}/200), \tag{25}$$

we use a Kalman likelihood whose measurement mean and variance are the exact first two moments of $\log(\chi^2_{200}/200)$; only its Gaussian shape is an approximation. The weekly fit gives

$(\rho, \phi, v_g) = (0.872, 0.249, 0.0556)$, and the daily+weekly fit gives $(0.878, 0.139, 0.0649)$. On the 158 2025 innovations, log-radius negative log likelihood falls from 1.986 to 0.089 in the weekly model and from 2.453 to 0.125 in the daily+weekly model relative to a pre-2025-scaled Gaussian benchmark.

For comparison we also fit variance-normalized multivariate Student-t innovations, either IID or with a persistent score-driven common scale. The fitted degrees of freedom are 42.6 and 34.5. On the same 2025 innovations the score-driven law reduces full-vector negative log likelihood per dimension from 1.377 to 1.364 in the weekly geometry and from 1.351 to 1.332 in the daily+weekly geometry; the improvement over IID Student-t is negligible.

Table 4 reports the probability of exceeding the pre-2025 empirical 95% radius at each horizon. "Market" is the realized 2025 rate; every simulated estimate uses one million paths.

Geometry	Innovation law	1W	2W	3W
Weekly	2025 market	5.063%	5.696%	5.696%
	Gaussian	0.1470%	0.00439%	0.00539%
	IID Student-t	4.350%	1.275%	1.164%
	Score-driven Student-t	4.523%	1.542%	1.435%
	Log-variance diffusion	3.960%	1.153%	1.032%
	Independent historical replay	5.143%	2.444%	2.386%
Daily+weekly	2025 market	6.329%	6.329%	5.696%
	Gaussian	0.0318%	0.000140%	0.000133%
	IID Student-t	4.139%	0.986%	0.856%
	Score-driven Student-t	4.119%	1.071%	0.931%
	Log-variance diffusion	3.751%	0.778%	0.659%
	Independent historical replay	5.150%	2.412%	2.318%

Table 4: Probability of exceeding the pre-2025 market 95% normalized-radius threshold. Market frequencies use 158 2025 transitions; simulation uses one million paths per law.

The no-replay diffusion is therefore a meaningful improvement, not a complete tail model. It also produces squared-radius serial correlations of 0.206 and 0.116, compared with 0.133 and 0.091 in the weekly and daily+weekly market histories; Gaussian and IID laws have zero population correlation. The 2025 sample is small and overlapping: 15-origin circular-block bootstrap 95% intervals for the market exceedance rates are approximately 0.6–14.6% across the six geometry/horizon cells. Differences of a percentage point should therefore not be over-interpreted. Independent historical replay is still closest in marginal tail mass, while the diffusion is a fully parametric generator with fresh support and explicit persistence.

4.3 Chronological surface-path comparison

Horizon radii test the latent innovation law but do not directly test paths of decoded option prices. We therefore form every available sequence of four fitted market-state surfaces separated by one full trading week, giving three consecutive weekly changes. Here "observed" means the date-specific latent state fitted to that day's irregular quote book and then passed through the common SANOS decoder; it is not a second interpolation of the raw quotes. Every candidate simulator starts from that fitted surface at the same origin date. Its path features are changes in normalized call prices on the Cartesian panel of target locations

$$T \in \{1, 5, 10, 20, 40, 70\} \text{ business days}, \qquad K/S \in \{0.85, 0.90, 0.95, 1.00, 1.05, 1.10, 1.15\}.$$

Each target expiry is mapped to the closest distinct model expiry. The daily+weekly geometry therefore has $3 \times 6 \times 7 = 126$ features; in the weekly geometry the 1D and 5D targets both map to 5D, leaving $3 \times 5 \times 7 = 105$. Coordinates are centered and scaled using pre-2025 observed paths only.

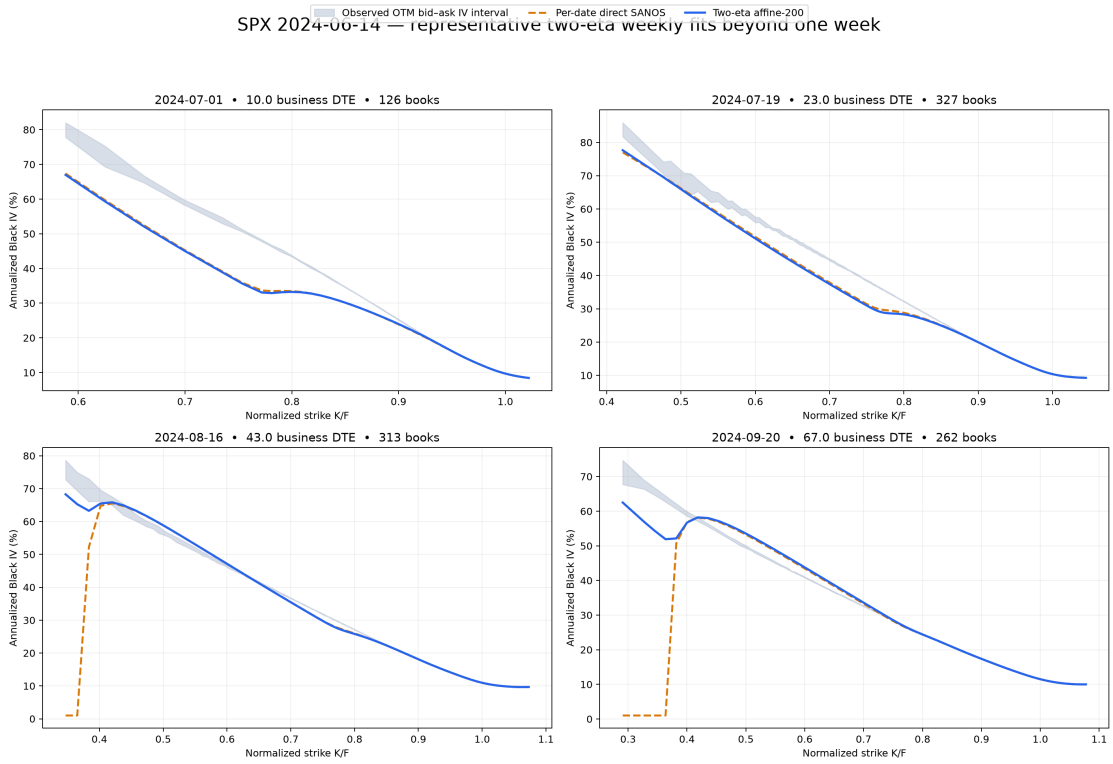


Figure 5: Representative SPX fit on 2024-06-14. The panels compare observed OTM bid–ask implied-volatility intervals, the per-date SANOS preprocessing fit, and the affine weekly model at four market expiries.

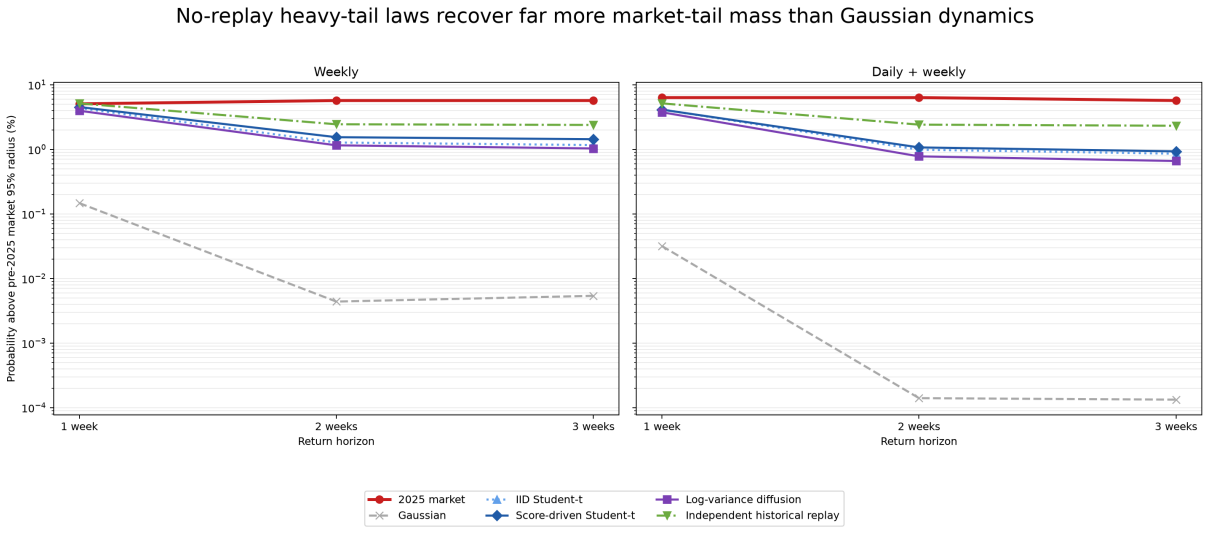


Figure 6: Probability of exceeding the pre-2025 market 95% normalized-radius threshold, on a logarithmic scale. Both no-replay heavy-tail laws recover orders of magnitude more tail mass than Gaussian innovations but remain too light at two and three weeks.

For this path-level test, the historical comparator replays an entire pre-2025 three-week sequence of fitted residuals, rather than drawing its three weekly residuals independently. It is therefore a deliberately strong finite-sample benchmark that preserves short-run clustering, but it is not a new parametric law and cannot generate innovations outside the stored blocks.

We report the standardized energy distance and an unbiased Gaussian-kernel maximum mean discrepancy (MMD) between observed and simulated 2025 paths. For the null distribution, entire paired chronological blocks of 15 origin dates swap their observed/simulated labels; the block length matches the three-week feature horizon and retains most overlap-induced dependence. 1,999 random block swaps give a Monte Carlo p-value with minimum positive resolution 0.0005.

The 2025 split is a chronological holdout for fitting any auxiliary tail law, constructing a historical-residual pool, and estimating the feature standardization. The shared affine decoder and the baseline Gaussian AR law are nevertheless taken from the all-history checkpoint. The path comparison is therefore a holdout diagnostic for the innovation-law extensions conditional on the learned representation, not a fully frozen pre-2025 model forecast.

Table 5 reports 144 paired 2025 paths for each geometry. The p-value is the block-swap test for MMD; energy distance is reported as a second scale-sensitive discrepancy. None of the five candidates is rejected at 5%. This is evidence against a gross path-distribution failure, but not a ranking theorem: IID Student-t has the smallest weekly energy distance, whereas Gaussian has the smallest daily+weekly energy and MMD. The log-variance diffusion remains statistically plausible while improving the separate tail and persistence diagnostics above.

	Weekly			Daily+weekly		
Innovation law	Energy	$10^3\times\text{MMD}^2$	p	Energy	$10^3\times\text{MMD}^2$	p
Gaussian	0.572	4.602	0.550	0.810	1.594	0.704
IID Student-t	0.540	5.016	0.344	0.854	3.788	0.457
Score-driven Student-t	0.569	4.389	0.225	0.838	3.786	0.416
Log-variance diffusion	0.606	5.476	0.378	0.854	3.480	0.415
Historical block replay	0.592	6.671	0.328	0.857	6.968	0.345

Table 5: Chronological decoded-surface path diagnostics on 144 paired 2025 origins. Each path contains three consecutive weekly changes; p is based on 1,999 paired 15-origin block swaps.

4.4 Stability diagnostics

Statistic	Weekly	Daily+weekly
State dimension	200	200
Dates	891	891
Exact 1W innovations	861	861
$\rho(A)$	0.9577	0.9561
Slowest-mode half-life	16.1 weeks	15.4 weeks
Minimum real eigenvalue of κ	0.0432	0.0449
Invariant covariance condition number	3.51	3.64
Invariant covariance effective rank	196.5	196.4

Table 6: Fitted state-law diagnostics.

Both laws are stable and retain almost full effective rank in state space. A simulator can therefore start from an observed state or draw $h_0 \sim \mathcal{N}(m, P)$ when a stationary initialization is appropriate.

4.5 Optional conditional spot block

Remark 4.1 (Normalization with forwards and discount factors)**.** *If the market has discount factors* $\boldsymbol{DF}_t(T) = \exp(-R_t(T))$ *and forwards* $\boldsymbol{F}_t(T) = \boldsymbol{S}_t \exp(R_t(T) - D_t(T))$*, the normalized one-step log-spot return is*

$$\Delta s_{t+1} = \log \boldsymbol{S}_{t+1} - \log\big(\boldsymbol{DF}_t(\Delta t)\boldsymbol{F}_t(\Delta t)\big) = \log \boldsymbol{S}_{t+1} - \log \boldsymbol{S}_t + D_t(\Delta t).$$

Option prices require no further drift adjustment here because the input surface has already been expressed in forward- and discount-normalized units.

The normalized option surface can be simulated autonomously. When a cash spot path is required, log-spot $s_t = \log S_t$ is attached after the surface fit. A simple conditional regression is

$$\Delta s_{n+1} = \mu^s + \beta' Q^{-1/2} \varepsilon_{n+1} + \varsigma \eta_{n+1}, \qquad \eta_{n+1} \sim \mathcal{N}(0,1), \tag{26}$$

with η independent of the surface innovation. This preserves the autonomous, stationary surface law while reproducing contemporaneous spot–surface dependence. Log-spot itself is not included in the invariant surface distribution.

4.6 Decoded implied-volatility geometry

Although h_t has 200 coordinates, decoded implied-volatility variation is much lower-dimensional. We differentiate the affine SANOS decoder at the invariant mean and combine its Jacobian with P. The resulting PCs are directions in decoded implied-volatility space, not the learned state coordinates.

Cumulative decoded-IV variance	Weekly	Daily+weekly
PC1	74.85%	77.68%
PC1–2	86.97%	86.98%
PC1–4	95.63%	94.50%
PC1–8	99.46%	99.39%

Table 7: Decoded implied-volatility variance under the invariant state law.

Figure 7 gives the full leading spectrum. PC1 is a nearly parallel volatility-level move and explains 74.85% of weekly decoded variance and 77.68% of daily+weekly decoded variance. Four decoded-IV factors explain 95.63% and 94.50%, respectively. This does not imply that four latent coordinates would fit the quote book: the PCA is performed *after* the 200-state model has been passed through the nonlinear SANOS decoder and weighted by its invariant covariance.

The invariant means in Figure 8 are the natural "base volatility" surfaces of the two statistical models. Their broad term-structure and skew geometry is similar. The difference is concentrated at the front: the daily+weekly model has four additional expiry slices with independent DLV and forward-volatility coordinates, whereas the weekly model extrapolates from 5D.

Figures 9 and 10 show the first eight eigensurfaces. Each panel is one invariant standard deviation and has its own vertical scale; PCA signs are arbitrary. PC1 is the global level. PC2 is primarily a broad skew/term-structure rotation, while PC3 couples skew and maturity. Higher modes become increasingly localized. The daily+weekly model places visibly more of those localized modes at 1D–4D, which is precisely the flexibility needed to fit the front-end quote book.

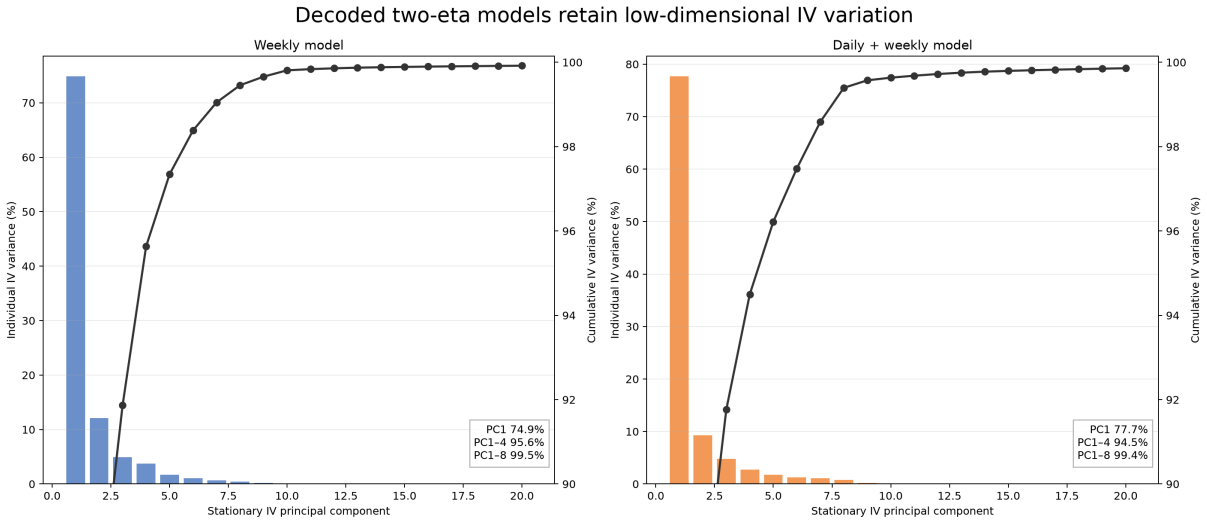


Figure 7: Variance explained by PCA of decoded implied-volatility levels under the fitted invariant state laws. The 200 affine coordinates are not themselves these PCs.

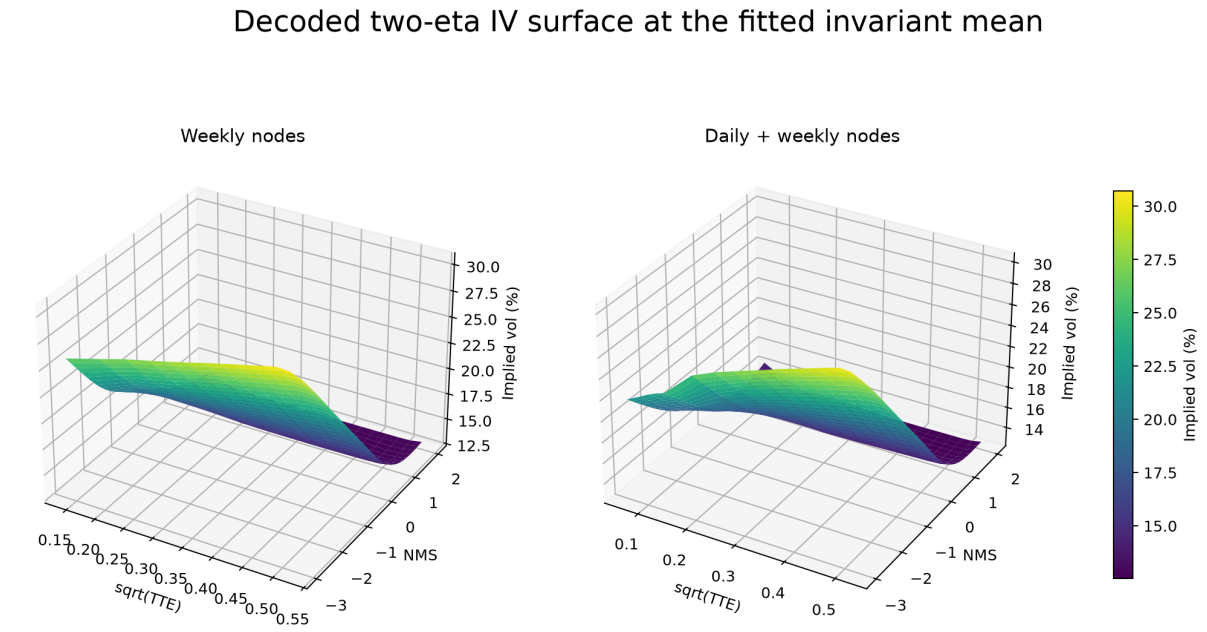


Figure 8: Base implied-volatility surfaces, obtained by evaluating each affine SANOS decoder at the invariant state mean. Axes are square-root time-to-expiry, normalized moneyness, and annualized implied volatility.

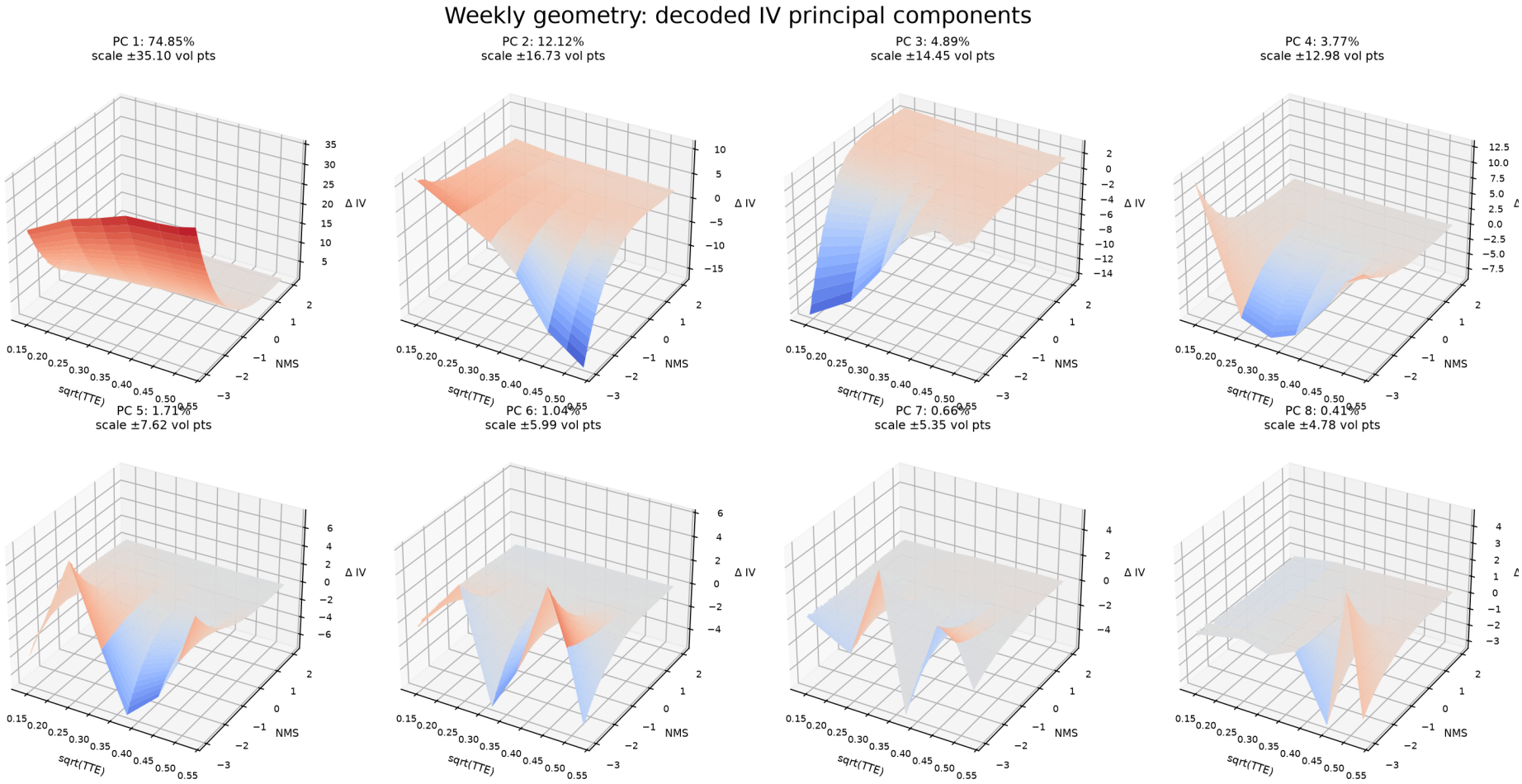


Figure 9: Weekly geometry: decoded implied-volatility PCs 1–8 at one invariant standard deviation. The title of each panel reports explained variance and the panel-specific move in volatility points.

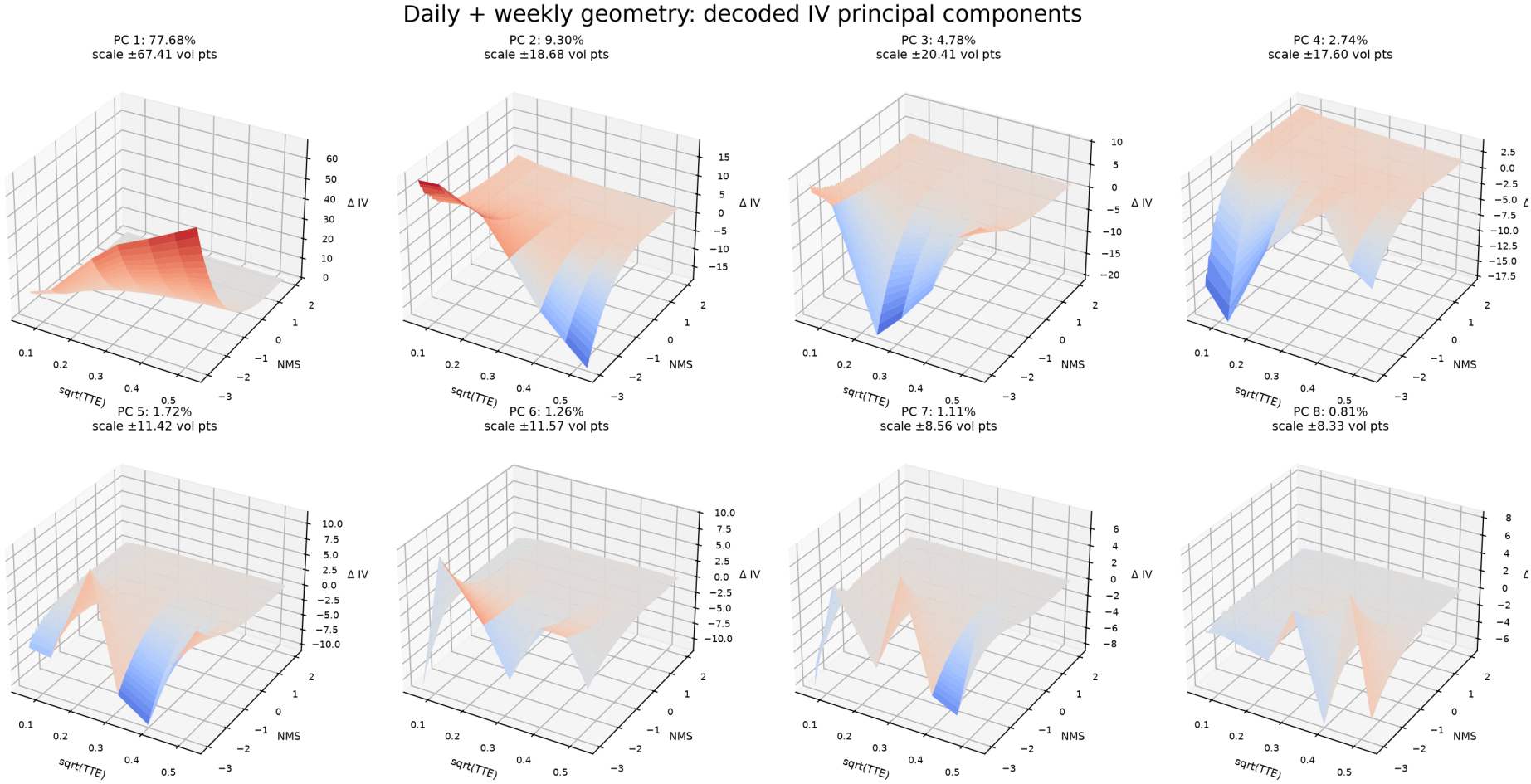


Figure 10: Daily+weekly geometry: decoded implied-volatility PCs 1–8 at one invariant standard deviation. Additional front-end modes appear because 1D–4D are explicit model expiries.

4.6.1 Invariant-distribution visualization

The PCA summarizes local covariance at the invariant mean; complete nonlinear decodes verify what that covariance produces as a surface distribution. Figure 11 shows six independent draws from each fitted $\mathcal{N}(m, P)$ state law. The weekly model's sub-5D ATM term structure is visibly constrained by extrapolation from its first node. The daily+weekly model can generate sharp but still statically admissible front-end dislocations.

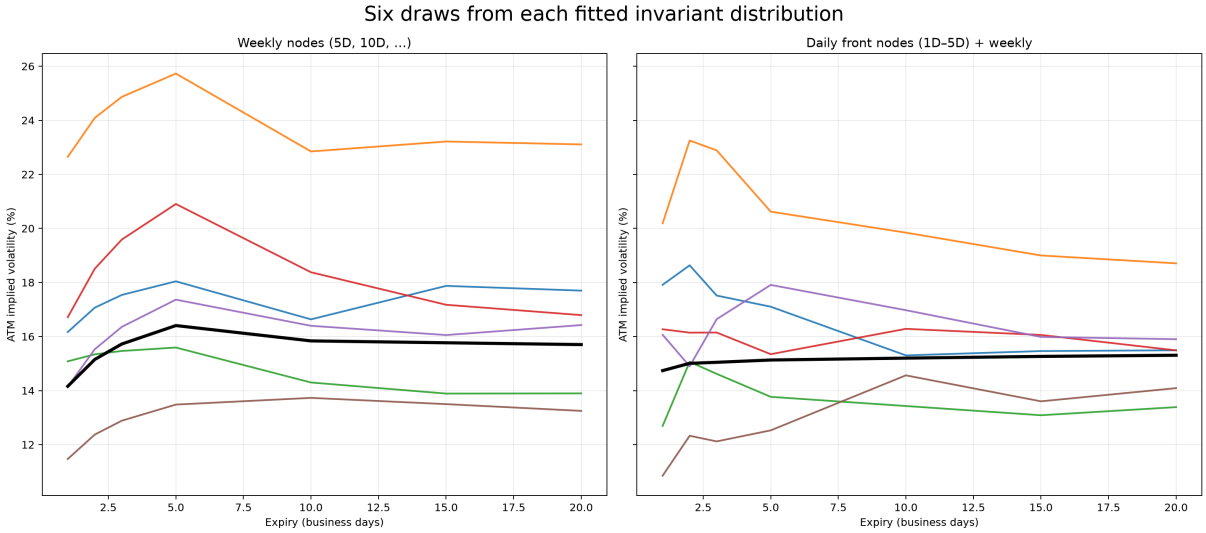


Figure 11: ATM implied-volatility term structures from six independent invariant-state draws for each model. Black is the invariant-mean decode. These are stationary cross-sectional samples, not consecutive time steps.

Figure 12 uses the same 200-dimensional standard-normal seed for both models, mapped through each model's own invariant covariance and decoder. It provides a controlled visual comparison of complete draws: their long-end shapes remain broadly comparable, while the daily+weekly surface has materially richer short-end geometry.

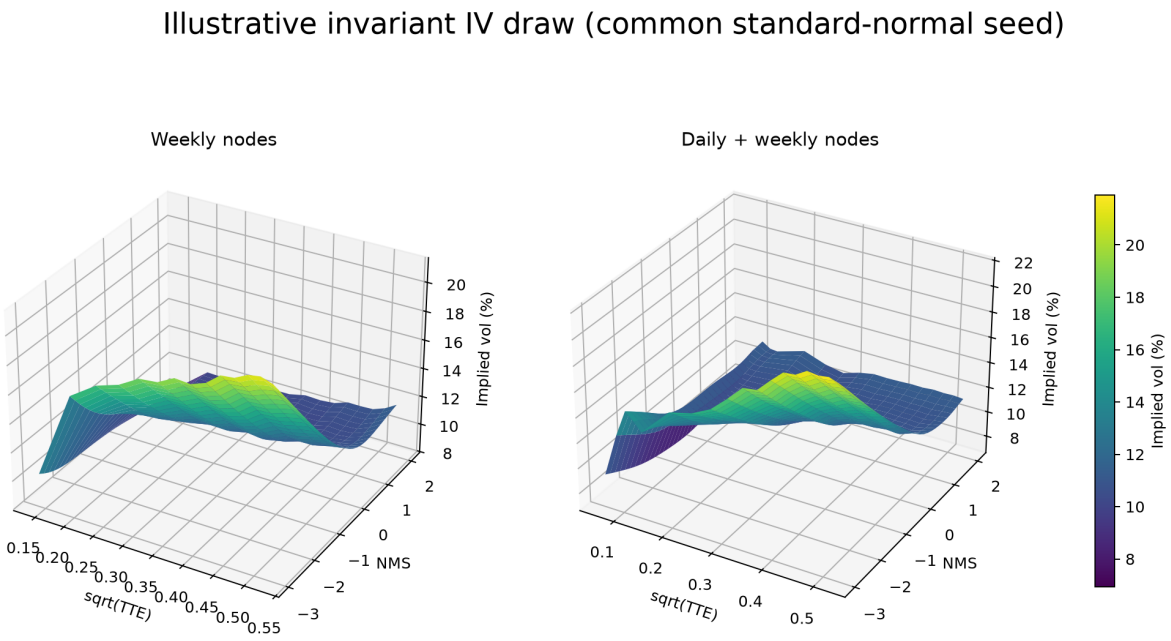


Figure 12: Paired complete invariant-state surface draws. A common standard normal seed is transformed by each fitted invariant covariance before being decoded by the corresponding affine SANOS model.

4.7 Simulation algorithm

The statistical DYSANOS simulator is summarized as follows.

1. Choose the daily+weekly geometry and initialize h_0 from an observed date or from the invariant state distribution.
2. Propagate the state with the exact Gaussian transition (21), or use the no-replay log-variance diffusion (23)–(24) when persistent tails are required. Historical innovation replay remains a non-parametric weekly benchmark.
3. If cash prices are required, propagate s_t conditionally using (26).
4. Form $u_t = b + Bh_t$, map it to bounded SANOS parameters through (11) and (15), and apply the density recursion of Section 3.
5. Evaluate $C_t(T, K)$ at any desired strike and expiry. For a fixed cash contract (τ, k) use $T = \tau - t$ and $K = k/S_t$.

The AR transition and SANOS decoder serve different roles. Stability of the first supplies a coherent long-run statistical state process; the second guarantees a valid option cross-section for every realized state. The combination is what we mean by consistent $\mathbb{P}$-simulation in this paper.

4.8 One-session hedging diagnostics

We assess whether the fitted conditional distributions are useful for a concrete downstream task. Let Z_{t+1} denote the normalized one-session change of a signed option position and let $X_{t+1} = S_{t+1}/S_t - 1$. At the close of date t we choose one spot position a_t by each of three criteria. Mean–variance hedging solves

$$\min_{a_t} \mathrm{Var}_t(Z_{t+1} + a_t X_{t+1}). \tag{27}$$

The two optimized-certainty-equivalent (OCE) hedges solve

$$\sup_{a_t, y_t} \left\{ y_t + \mathbb{E}_t \left[u_\lambda(Z_{t+1} + a_t X_{t+1} - y_t) \right] \right\}, \tag{28}$$

using either entropy utility $u_\lambda(x) = -e^{-\lambda x}/\lambda$ or the power-exponential utility

$$u_\lambda(x) = \begin{cases} (1 - e^{-\lambda x})/\lambda, & x \geq 0, \\ x - \lambda x^2/2, & x < 0. \end{cases} \tag{29}$$

We use $\lambda = 50$ and optimize the cash translation y_t jointly with the hedge. Reported first-order conditions are evaluated on the same scenario set used for optimization.

The first experiment hedges the closest-to-ATM European PM call and put with one business day to expiry. At each close t, its scenario set contains all adjacent full-session spot returns whose endpoint is already observed by t; the forward changes are sample-centered during hedge estimation. The next retained close is used as a settlement proxy. We report buyer and seller positions separately because the OCE hedge need not be antisymmetric.

The second experiment is an open-interest portfolio proxy. We treat the date-t open-interest field as an after-close observation and assign it only to the next one-session experiment; its publication timing is not independently verified. Since open interest is unsigned, we transparently assume that a dealer is short every retained call and put and normalize the positions to one unit of gross open interest. This is not an estimate of net dealer inventory. Only matched European PM contracts with 2–70 business days to expiry and $K/S_t \in [0.75, 1.25]$ enter. Scenario changes pair one-session state innovations whose endpoints are observed by close t (including

the transition ending at t) with their contemporaneous spot returns, while realized portfolio changes use matched market mid-prices at the next close. The 2025 window supplies 159 such close-to-next-close observations without a missing-book failure. Each proxy contains 4,058–8,547 matched contracts (median 7,009); median gross open interest is 2.82 million contracts.

For both experiments we compare the chronological unhedged and hedged P&Ls by root-mean-square loss, mean absolute loss, and lower-tail summaries. A paired circular block bootstrap with five-session blocks reports 95% percentile intervals. A centered block-bootstrap null reports one-sided p-values for reductions in absolute and squared loss. Option marks are bid–ask midpoints and spot-hedge transaction costs are omitted, so these diagnostics isolate hedge quality rather than executable net profitability. The fitted all-history decoder is not a frozen pre-2025 forecasting model, so these are historical hedge evaluations rather than claims of fully untouched out-of-sample performance.

ATM one-day results. Table 8 reports normalized root-mean-square P&L in basis points of initial spot over 158 close-to-next-close observations in 2025. All three criteria reduce RMS P&L by 45–48%. Their differences are small: entropy OCE is best for both buyer books, while mean–variance or power-exponential OCE is marginally better for the seller books.

Signed ATM 1D position	Unhedged	Mean–variance	Entropy OCE	Power-exp OCE
Buyer call	84.79	44.67	**44.07**	44.12
Seller call	84.79	**44.67**	45.41	44.72
Buyer put	81.11	44.11	**43.56**	43.61
Seller put	81.11	**44.11**	44.76	44.16

Table 8: RMS one-session P&L in basis points of initial spot for the ATM one-day option experiment. The paired five-session block-bootstrap p-value for absolute-loss improvement is 0.0001 for every hedge. For squared loss it is approximately 0.078 for calls and 0.032 for puts. Across methods, the 95% block-bootstrap intervals for the mean absolute-loss reduction are 12.2–28.1 basis points for calls and 17.7–32.7 basis points for puts.

Across the four signed books, all 1,896 hedge estimates are finite, no spot bound is active, and every numerical OCE optimization reports convergence. The maximum absolute first-order residual is 3.35×10^{-7}. The chronological scenario sets use only transitions observed by close t; nevertheless, the option selection and evaluation period remain a historical diagnostic rather than a fully frozen model forecast.

Open-interest portfolio proxy. Table 9 reports the same diagnostics for the 159 portfolio proxies. The date-t after-close OI field is used as a proxy for the book held from close t to close $t+1$, exactly as stated above; this convention does not assert that the field was tradable information before the date-t close. Mean–variance and both OCE criteria again produce nearly identical hedges. RMS loss falls by about 49% and mean absolute loss by about 54%.

Method	RMS	Mean absolute	Lower 5% mean	$p_{\lvert\cdot\rvert}$	$p_{(\cdot)^2}$
Unhedged	28.02	10.25	−65.35	–	–
Mean–variance	14.35	4.71	−40.85	0.0079	0.0820
Entropy OCE	14.43	4.72	−40.93	0.0076	0.0822
Power-exp OCE	14.36	4.71	−40.85	0.0078	0.0820

Table 9: One-session P&L of the open-interest portfolio proxy, in basis points of gross open interest. The final two columns are one-sided centered five-session block-bootstrap p-values for reductions in absolute and squared loss relative to the unhedged proxy.

The 95% block-bootstrap interval for the mean absolute-loss reduction is 2.34–9.81 basis points across the three methods. The squared-loss test is only borderline at conventional levels, so the reduction in RMS should not be described as statistically definitive. All 477 fitted hedge problems are finite, no position bound is active, and no optimizer reports failure; the largest absolute first-order residual is 1.58×10^{-7}. These results measure the usefulness of the conditional scenario law for the stated proxy, not the performance of an observed dealer inventory.

4.9 Scope and limitations

The 80.92% market-fit figure is computed on the population used to estimate the final all-history model. It is not a claim of 80% future forecast coverage. A formal forecasting study should freeze the decoder and dynamics before the evaluation period.

The Gaussian state law severely understates tail frequency. The no-replay log-variance diffusion improves likelihood, tail mass, and serial-tail dependence, but still underproduces two- and three-week extremes. Historical innovation replay is closer in marginal tail mass, while the path-level tests have only 144 overlapping 2025 origins and do not distinguish the candidates sharply. A richer multiscale or regime-dependent volatility law is the most direct next extension.

Finally, static arbitrage-freedom is a property of each decoded surface. It does not by itself impose restrictions on expected returns of fixed contracts through simulated time. This paper deliberately evaluates a statistical historical law and makes no additional dynamic pricing assertion.

5 Conclusion

We presented with DYSANOS the first generative option market model built from smooth, strictly statically arbitrage-free SANOS surfaces. Our approach does not rely on approximate static constraints; each smooth option surface is strictly arbitrage free and allows pricing of any option on GPU. We have shown how to leverage our representation to build a simple, but already fairly powerful baseline model which can generate spot and option paths for many days in the future up to years. The surface-first AR(1) captures first-order level, skew, term, and leverage effects. Replacing expiry-specific smoothing coordinates by a front and a term coordinate leaves the reported quote fit essentially unchanged, and the daily+weekly affine model places 80.92% of the full quote population inside bid–ask. A no-replay log-variance diffusion improves the deficient Gaussian tail law and reproduces tail persistence, although multi-week tail mass remains too low. The one-session ATM and open-interest proxy experiments show that the fitted conditional distribution can materially reduce historical hedging loss under three different criteria.

We have provided a baseline model to generate long paths of daily spot and option prices. Each option surface is free of static arbitrage. Moreover, we can price *any* option on each path given the ML-SANOS parameters for the respective sample and time step. Contrary to previous results, agents can therefore choose which options to trade at each time step.

6 Thanks

We want to thank the team at Wharton for their WRDS https://wrds-www.wharton.upenn.edu portal and Option Metrics https://optionmetrics.com/united-states for providing their IvyDB data via this service. Without this data our work would not have been possible.

A Appendix: Data Pipeline

In the interest of full transparency we here discuss main elements of the data pipeline used in our experiments.

The market-data pipeline constructs a standardized daily time series of SPX option surfaces. Raw option quotes are cleaned, normalized, filtered, checked against an arbitrage-free reference surface, and finally collected into irregular quote books suitable for direct model training.

The pipeline processes SPX option data between January 2nd, 2020 and August 29th, 2025. The latter is the latest day available at the time of writing. Only valid full-day NYSE trading days are considered.

A.1 Pipeline Overview

The complete transformation can be summarized as

$$\begin{aligned}\text{raw SPX option quotes} &\longrightarrow \text{daily cleaning and enrichment}\\ &\longrightarrow \text{arbitrage-free quote surface using linear SANOS}\\ &\longrightarrow \text{liquidity and moneyness filters}\\ &\longrightarrow \text{shared 200-strike SANOS model geometry}\\ &\longrightarrow \text{direct native and affine-200 quote fits.}\end{aligned}$$

The key insight when working with Option Metrics IvyDB data is that the forward curves from the `Forward_Price` table do not line up with the option data themselves. We therefore compute our own forward from weighted put-call parities. We also convert the tightest options into call prices. Such data still has arbitrage, hence we use a linear SANOS pre-processing step to obtain fitted prices within bid/ask which adhere to basic linear no-arbitrage conditions (the absence of which would indicate actionable arbitrage). We drop any strikes where such a fit cannot be achieved, assuming that fit issues are due to data alignment challenges rather than genuine trading opportunities.

A.2 Raw Daily Market Data

For every eligible date d, the underlying data contain quantities such as

- the SPX spot price S_d and previous spot price S_{d-1};
- option strikes and expiration dates; call and put bid and ask quotes; volume and open interest; AM/PM settlement indicators;
- forward and discount-curve information (with forwards being re-calculated below).

The default raw-data cleaning removes

- same-day expirations;
- options with bids below 0.011 (the minimum tick size is 0.05\$ for series trading below 3\$);
- observations with missing or invalid quotes; observations with an invalid settlement flag; irregular expirations; dates or ticker-date combinations known to contain bad data (e.g. 2025-01-08 which was a holiday).

A.3 Enrichment and Expiry Filtering

The cleaned quotes are enriched with

- business days and year fractions to expiry where AM options are counted as 20% of the day;

- implied forward prices computed off put-call parity per expiry for all strikes for which both call and put were traded and have valid bid/ask prices and sizes, weighted by put-call implied forward bid-ask spread.[6]
- using the new forward, define "market" mid call prices between the actual call prices and synthetic call prices from the puts;
- compute implied volatilities; forward- and discount-adjusted prices, strikes, and Greeks.

An expiration is retained only if it satisfies the default liquidity and quality requirements. In particular,

- at least 10 strikes must have either a traded call or put;
- at least 4 strikes must have both a traded call and put;
- the discounted forward normalized spread must be at least 10^{-6};
- implied volatility must lie in
$$0.01 \leq \sigma^{\text{impl}} \leq 5.$$

A.4 Arbitrage-Free Preprocessing

Before the final option filter is applied, an arbitrage-free representation of the daily market is constructed using [BHK+26]. The default configuration uses surface mode and performs the following operations:

1. Extend the strike grid to improve boundary treatment.
2. Fit a linear (DLV) no-arbitrage option-price curve separately for each expiry.
3. Drop expiries for which a valid fit cannot be obtained.[7]
4. Remove locally redundant strikes associated with negligible fitted density.
5. Fit a surface across strike and expiry to control both butterfly and calendar arbitrage.

At least five strikes per expiry are required. The default minimum fitted density is 10^{-4}, and the minimum admissible ATM volatility is 0.05.

A.5 Option Filtering

The default option filter retains observations satisfying

$$\text{BDTE} \leq 3 \times 252 = 756,$$

where BDTE denotes business days to expiry.

The normalized-moneyness filter is

$$-5 \leq z_{\text{ATM}} \leq 2,$$

where z_{ATM} is normalized moneyness computed using the ATM volatility.

[6] We compute $\text{fwd}^{\text{ask}}(K) := +\text{Call}^{\text{ask}} - \text{Put}^{\text{bid}} - \text{df}K$ and $\text{fwd}^{\text{bid}}(K) := +\text{Call}^{\text{bid}} - \text{Put}^{\text{ask}} - \text{df}K$ using the discount factor sourced from IvyDB. Then we define $\text{fwd}^{\text{mid}}(K) := 0.5\,(\text{fwd}^{\text{ask}}(K) + \text{fwd}^{\text{bid}}(K))$ and $\gamma^{\text{mid}}(K) := 0.5\,(\text{fwd}^{\text{ask}}(K) - \text{fwd}^{\text{bid}}(K))$ and
$$\text{fwd} := \frac{\sum_K \gamma(K)^{-1} \text{fwd}^{\text{mid}}(K)}{\sum_K \gamma(K)^{-1}}$$

[7] Existence of such a fit is equivalent to the market not having actionable arbitrage using bid/asks as quoted.

Additional requirements are

$$\begin{aligned} \text{VegaSqrtVar} &\geq 10^{-3}, \\ \log(\text{volume}) &\geq 1, \\ \log(\text{open interest}) &\geq 1. \end{aligned}$$

Here, "VegaSqrtVar" refers to Vega divided by sqrt of expiry. It is therefore the sensitivity to total volatility of the option.

Each retained expiry must contain at least 10 options. At most 1000 options are kept per day. If more than 1000 observations survive, the filter first reserves observations around the money for every expiry and then fills the remaining capacity using increasing absolute normalized moneyness.

It should be noted that this filter removes many options post 1Y for the early years of the training period due to the minimum volume traded filter. We suspect that this information in IvyDB is not reliable in the past. Future tests might want to remove the volume and/or open interest filters. (It is an option in our data pipeline parametrization.)

A.6 Spot Normalization

Price-like quantities are normalized by the current spot S_d to obtain variables of scale around 1. For example,

$$\widetilde{K}_d^i = \frac{K_d^i}{S_d}, \qquad \widetilde{C}_d^i = \frac{C_d^i}{S_d}, \qquad \widetilde{F}_d^i = \frac{F_d^i}{S_d}.$$

The original spot is retained as $\text{spot_ref}_d = S_d$. (The normalized log-spot level is passed on as h^0 later.)

A.7 Arbitrage-free preprocessing and model geometry

The preceding steps produce tradable normalized calls and puts with their observed bid–ask intervals. A per-date arbitrage-free SANOS fit supplies a smooth reference surface, an ATM-volatility curve, and an availability check for the retained quotes. The completed Direct-200 experiments do *not* resample this information onto a secondary reference grid. Instead, every eligible original quote enters the loss at its observed strike and expiry.

The decoder uses the two model-expiry geometries stated in Section 3.1.2: 14 weekly expiries from 5D to 70D, or the same weekly grid augmented by 1D–4D for a total of 18 expiries. On each date it builds one shared support with $N = 200$ inner strikes. Half of the normalized-moneyness candidates cover the first model expiry and half cover the final model expiry over $z \in (-3, 2)$. The candidates are merged and spaced, and the two boundary strikes are

$$\hat{K}_t^0 = \tfrac{1}{2}\hat{K}_t^1, \qquad \hat{K}_t^{N+1} = \tfrac{3}{2}\hat{K}_t^N.$$

The same support is used at every model expiry on that date; only the density and DLV coordinates carry an expiry superscript.

A.8 Fitting ML-SANOS

The native control first fits one unrestricted vector $u_t \in \mathbb{R}^{M(N+1)+2}$ to each date's irregular quote book. The completed affine model then uses the map

$$h_t \in \mathbb{R}^{200} \longmapsto u_t = b + Bh_t \in \mathbb{R}^{M(N+1)+2}$$

from (14). It is initialized from the native controls and then fitted jointly across all 891 dates together with the stable weekly state dynamics. Prices are evaluated directly at every observed quote; the training weights and inside-spread objective are those in (16).

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